

Sessional 1 (Nov (2022))
B.Tech (ME) - 5th Sem
HMT (PCC-ME-502-21)

Duration: 90 Minutes

Max. Marks: 30

Instructions:

- Attempt all questions
- All questions carry equal marks

- Q1** A horizontal steel pipe having a diameter of 5 cm is maintained at a temperature of 50°C in a large room where the air and wall temperature is at 20°C . The surface emissivity of the steel may be taken as 0.8. Calculate the total heat lost by the pipe per unit length. Heat-transfer coefficient for free convection with this geometry and air is $h = 6.5 \text{ W/m}^2\cdot^{\circ}\text{C}$. (10)
- Q2** What are the modes of heat transfer? Discuss in detail with suitable examples. One face of a copper plate 3 cm thick is maintained at 400°C , and the other face is maintained at 100°C . How much heat is transferred through the plate? The thermal conductivity for copper is $370 \text{ W/m}\cdot^{\circ}\text{C}$ at 250°C . (10)
- Q3** What are radiations? How they are generated? What are the different types of radiation? Also, discuss the basic working of solar photovoltaic cells with suitable data and diagrams. (10)



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